

AI Email Triage & Auto-Draft Assistant

An n8n automation that watches a Gmail inbox, classifies every incoming email with AI, applies structured labels, alerts the team on Telegram, and drafts a reply — without ever sending on its own.

n8n

Groq LLM

Gmail

Telegram

1 Executive Summary

Business inboxes mix invoices, complaints, job applications and routine noise, and sorting it manually costs time and delays the messages that actually matter. This workflow reads every new email the moment it arrives, classifies it by department, type and priority, labels it in Gmail, pings the team on Telegram for anything urgent, and drafts a reply when one is useful — leaving the final send decision to a human.

2 Key Metrics

Trigger	Gmail Trigger, polling the inbox every minute
Classification Engine	Groq-hosted GPT-OSS-20B, structured JSON output
Organisation Method	Nested Gmail labels (Department / Type / Priority / Needs Review)
Urgent Alerts	Telegram message with sender, category, reason and a direct Gmail link
Reply Handling	Draft created in Gmail Drafts only — never auto-sent
Confidence Review	A classification below 0.75 confidence is auto-flagged Needs Review
Safeguards	Untrusted-input prompt guarding, enum validation, no autonomous sending

1 min

INBOX POLL INTERVAL

0.75

REVIEW CONFIDENCE THRESHOLD

0

EMAILS AUTO-SENT

3 Business Value

- ✓ **Inbox zero, automatically sorted** — every email is filed under a clear department/type/priority label the moment it lands.
- ✓ **Nothing urgent gets missed** — Telegram fires instantly for deadlines, outages, complaints or payment issues.
- ✓ **Faster replies without losing control** — a draft is ready to review and send, but the AI never sends on its own.
- ✓ **Self-auditing** — low-confidence classifications are flagged Needs Review instead of silently trusted.

4 Architecture Overview

A one-time setup branch creates the label set; a continuous branch processes every new email.

FLOW	TRIGGER	PURPOSE
Label Setup (once)	Manual trigger, run one time	Creates the full set of nested Gmail labels the workflow will apply
Email Triage (continuous)	Gmail Trigger, every minute	Reads, classifies, labels, alerts on and drafts replies for each new email

5 How It Works

Label Setup	Continuous Triage
1 A code node builds the full list of nested labels: Department, Type, Priority, and Needs Review. 2 Each label is created via the Gmail API under an AI/ prefix, run manually once per account.	1 A Gmail Trigger polls every minute for new mail, spam and trash excluded. 2 Sender, body and a direct thread link are extracted and cleaned into a payload. 3 Groq classifies department, type and priority and decides if a reply is worth drafting. 4 Any value outside the allowed list falls back to a safe default; confidence below 0.75 adds Needs Review. 5 Labels are applied in one API call; urgent emails trigger a Telegram alert, useful replies get a Gmail draft.

6 Reliability & Security Design

- ◆ **Prompt-injection resistance** — the email body is treated as untrusted data; the model never follows instructions embedded inside it.
- ◆ **No autonomous sending** — the workflow can create a Gmail draft, but nothing in the flow ever sends an email.
- ◆ **Enum-safe parsing** — any classification outside the allowed list is discarded in favour of a safe default.
- ◆ **Confidence-based review** — classifications below 0.75 confidence are automatically flagged Needs Review.
- ◆ **Resilient AI calls** — the Groq request retries automatically on failure, reducing dropped emails.
- ◆ **Clear separation of duties** — classification, labelling, alerting and drafting are each their own auditable step.

7 Setup Checklist

- ☒ Connect Gmail OAuth2 (same account across every node), a Groq header-auth credential, and Telegram.
- ☒ Run “Run Label Setup Once” manually and confirm the AI/... labels appear in Gmail.
- ☒ Get your numeric Telegram Chat ID from @get_id_bot and paste it into the alert node.
- ☒ Test three scenarios: routine, time-sensitive, and reply-needed emails.
- ☒ Once all three pass, save and activate the workflow.

8 Node-by-Node Map

One-time Setup	Run Label Setup Once → Required Gmail Labels → Create Gmail Labels
Intake & Classify	New Gmail Email → Prepare Email → Classify with Groq → Validate AI Output
Labelling	Get Label IDs → Resolve Label IDs → Apply AI Gmail Labels
Action Branches	Is Urgent? → Send Telegram Alert Draft Needed? → Create Gmail Draft

SUMMARY

Designed and built end-to-end in n8n, demonstrating event-driven email intake, guarded LLM classification, structured Gmail organisation, and human-in-the-loop drafting.